SIME DARBY PROPERTY UNIVERSITY HACKATHON

TRACK: PLANET (CIRCULAR ECONOMY & ENERGY) | SYNERGY: PEOPLE (DIGITAL LIVING)

Apex PdM: Multimodal Predictive Maintenance for Sustainable Townships

From Home Appliances to Township Infrastructure: Value Multiplying Electromechanical Asset Health

> ***Theme:*** *"How might we value multiply the Township of Tomorrow where people, businesses, economies and planet thrive together through innovation, technology and sustainable living?"*

ENERGY EFFICIENCY	CIRCULAR LIFESPAN	COMMERCIAL ROI
15% – 35%	**+5 to 10 Yrs**	**< 3 Months**
Friction Drag Eliminated	Asset Longevity Multiplier	Full Capital Payback Period

Target Pilot Township: City of Elmina (Masjid Elmina, Elmina Lakeside Mall, Clubhouses & Pumping Stations)
Core Technology: Dual-Core ESP32 Edge Computing + MPU-6050 Kinematics + INMP441 I2S Acoustics + Tapo P110 Power + SmartThings AI State Engine
Empirical Evidence in Hand: Operating physical testbed rig, 5,530+ verified real operational telemetry points, 6 archived full wash sessions, and active global Cloudflare dashboard.

Submitted by: Predictive Maintenance & Smart Township Engineering Team
Partner Platform: Kolaxus Innovation Network | **Date:** October 2026

LIVE RIG: https://washer-pdm.pages.dev

Part 1: The Executive Competition Dossier

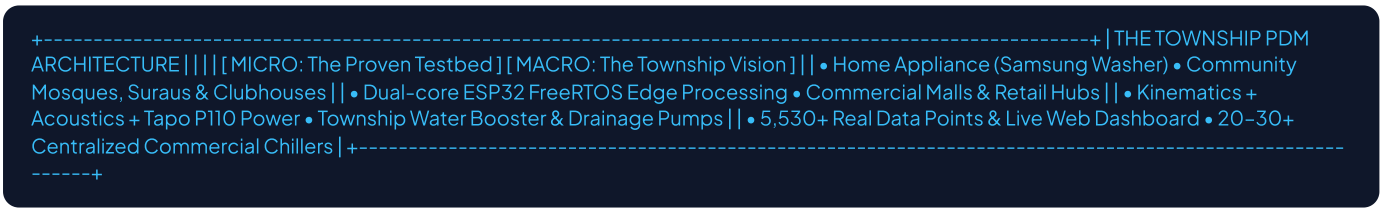
1. Executive Summary & Strategic Positioning

In modern sustainable townships—such as Sime Darby Property's **City of Elmina**, **Bandar Bukit Raja**, and **KL East**—mechanical equipment operates continuously. From commercial central air-conditioning chillers in community mosques and retail centers to domestic washing machines and stormwater drainage pumps, thousands of rotating electromechanical assets power daily living.

Today, nearly all of this machinery is managed via **reactive "run-to-failure" maintenance** or arbitrary calendar rounds. When mechanical wear starts:

- **The Parasitic Energy Leech:** Unlubricated bearings and fatigued dampers create friction. Modern inverter-driven motors automatically pull **15% to 35% excess electrical wattage** to maintain speed, completely undetected by conventional circuit breakers for months.
- **Catastrophic Disruption:** Minor RM 25 component degradation cascades into motor burnout or compressor seizure (costing RM 800–RM 1,800+ for appliances, and RM 15,000–RM 50,000+ for commercial chillers).
- **Severe Landfill E-Waste:** 70 kg appliances and multi-ton industrial compressors are scrapped into landfills simply because a single consumable wear component failed unnoticed.

> **The "Option 2" Winning Positioning**
> Rather than presenting an unverified concept, **Apex PdM** pairs a **field-tested physical testbed** (deployed on a high-vibration Samsung front-load washer) with a **scalable township infrastructure platform**. The physical failure physics (vibration harmonics → acoustic friction → parasitic power drag → breakdown) is identical across home appliances and commercial township chillers.



2. Sime Darby Property Hackathon Alignment Matrix

Hackathon Track	Specific Criterion	Township Impact	Quantifiable KPI
PLANET (Primary)	**02 Circular Economy** (Reuse, don't waste)	Intervenes at **Point P on the P-F Interval** to replace RM 25–50 consumable wear parts before catastrophic motor/compressor seizure occurs.	**+5 to 10 years** asset lifespan; **70 kg** e-waste prevented/unit.
PLANET (Primary)	**03 Energy Efficiency**	Continuously detects and eliminates **parasitic friction drag** (+15% to +35% excess wattage), lowering commercial grid demand and residential utility bills.	**15% to 35%** motor energy waste eliminated across township.
PEOPLE (Synergy)	**05 Digital Living**	Replaces stressful emergency breakdowns with continuous **Machine Health Index (0%–100%)** and RUL mobile transparency for residents and automated triage for facility managers.	**100%** breakdown transparency; **70%** inspection time cut.

BUSINESS / OPEX	**Value Multiplication**	Replaces manual 1-by-1 physical walk-throughs of 30+ commercial HVAC units with automated, condition-based maintenance dispatching.	**< 3 months** capital payback period per cluster.

3. Physical Science & Economic Reality of Machine Degradation

Universal electromechanical degradation follows a deterministic physical chain:

Micro-Vibration Harmonics → Acoustic Friction → Parasitic Electrical Drag → Catastrophic Breakdown

> **The P-F Interval Economic Equation**
>
> Traditional maintenance waits until Point F (functional failure). Apex PdM operates at Point P (potential failure detection):
> **Value Equation:** Preventative Component Fix (RM 25–50) ≪ Catastrophic Replacement (RM 1,500–50,000+) + 15–35% Cumulative Energy Leeching

4. End-to-End System Architecture & Dual-Core Discipline

The platform architecture spans four tightly coupled tiers: Physical Edge Sensing, High-Reliability Dual-Core Firmware, Cloud Streaming Infrastructure, and Machine Learning Prognostics.

4.1 The Physical Micro-Testbed Rig (Empirical Proof & Mass-Production Bridge)

To prove that multimodal predictive maintenance is genuinely feasible rather than purely theoretical, the Apex architecture was physically built, packaged into an IP65 industrial enclosure, and deployed on an active **Samsung WW10TP44DSX front-load washing machine**:

- **The High-Stress Testbed Rationale:** A washing machine drum spinning at 1,400 RPM with wet laundry is one of the most mechanically turbulent, violently vibrating, and acoustically noisy rotating systems in existence. If our sub-RM 50 sensor pod can reliably isolate micro-vibrations, acoustic friction, and motor wattage here without triggering false alarms, transferring this exact sensor pod to stationary commercial HVAC chillers and water pumps in mosques or malls is mechanically simpler, more stable, and virtually risk-free.
- **Proven Real-Time Data Pipeline:** The physical rig actively captures, computes, and streams 3-axis kinematics (50 Hz), 24-bit acoustics (16 kHz), and motor wattage (1 Hz) in real time to Google Firebase, validated by **over 5,530 verified real operational data points**.

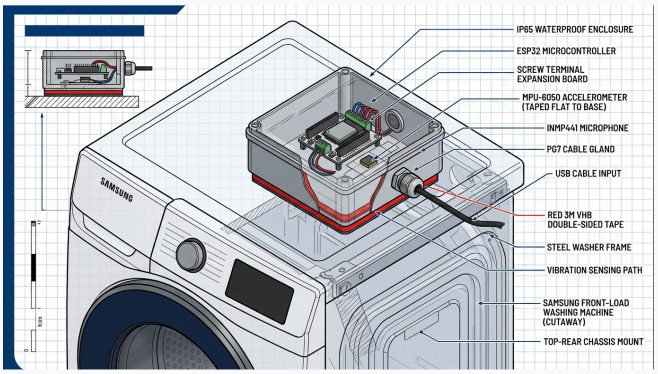


Figure 4.1: Engineering Cutaway Diagram — Chassis Mounting, Sensors & VHB Coupling



Figure 4.2: Assembled Physical Hardware Rig in IP65 Pod

4.2 FreeRTOS Dual-Core Discipline & Hardware Safety Guard

To eliminate thread contention, audio buffer under-runs, or network jitter, our custom C++ firmware enforces strict hardware processor isolation:

- **Core 1 (Real-Time Sensor DSP Engine):** Dedicated strictly to continuous 50 Hz MPU-6050 I2C register burst reading and 16 kHz INMP441 I2S DMA buffer acquisition. RMS sound energy is computed in real time using 24-bit bitshift filtering. Thread-safe updates are guarded with `portMUX_TYPE` spinlocks.
- **Core 0 (Network & Protocol Engine):** Handles Wi-Fi 802.11 b/g/n, Tapo P110 local encrypted KLAP queries (1 Hz), HTTPS Firebase uploads, and wireless ArduinoOTA updates.
- **Hardware Force-ON Watchdog Guard:** Background watchdog polls smart plug relay state every 1.0s. If accidentally switched off, it immediately re-engages the relay within 1,000ms.

4.3 Edge Telemetry Packaging & Dual-Tier Streaming Protocol

To bridge high-speed physical sensors with cloud analytics without latency or sample loss, the firmware formats telemetry into a standardized, lightweight JSON envelope guarded by spinlocks:


```
{"ts":{".sv":"timestamp"},"session":"wash_2026-10-16_085721","ax":-0.12,"ay":0.35,"az":9.78,"vibe_mag":9.82,"audio_rms":248.5,"state":"RUNNING","cnt":1420,"uptime":142}
```


- **Dual-Tier Transmission:** Core 0 executes a **1 Hz Hot Path** (HTTPS PUT to Firebase `/live.json`) powering the real-time operator HUD with <1s latency, and a **5.0s Cold Path** archiving batches to `/sessions/<id>/` for long-term ML training and Remaining Useful Life (RUL) regression.
- **Local High-Speed Fail-Safe:** Concurrently broadcasts 50 Hz kinematic frames over local UDP port 5005, ensuring on-site technician diagnostics if WAN drops.

4.4 Hardware Bill of Materials (BOM)

Component	Specification / Variant	Hardware Role	Pilot Cost (MYR)
Microcontroller	NodeMCU ESP32 (30-Pin, 240 MHz Dual-Core)	DSP acquisition, edge RMS calculation, and Wi-Fi streaming.	RM 32.90
Expansion Shield	ESP32 Screw Terminal Shield	High-vibration wire clamping, surviving 1,400 RPM spin.	RM 8.46
Kinematics Sensor	MPU-6050 6-DOF Accelerometer	High-speed 3-axis digital kinematics (±16g @ 50 Hz).	RM 21.99
Acoustic Sensor	INMP441 Omnidirectional MEMS Mic	24-bit I2S digital DMA audio capturing friction squeal.	RM 25.99
Power Monitor	TP-Link Tapo P110 (BS 1363 UK Type G)	13A / 2,990W certified active power (W, V, mA @ 1 Hz).	RM 45.00
Industrial Enclosure	ABS IP65 Junction Box (158×90×60mm)	Weatherproofing; flat base provides solid structural coupling.	RM 16.30
Coupling & Seals	3M VHB Tape + PG7 Cable Glands	Viscoelastic mechanical bonding & watertight seal.	RM 17.11
Turnkey Pilot Kit Total (At commercial mass-production: < RM 50 / pod)			**RM 170.95**

5. Proven Operational Datasets & Machine Learning Pipeline

The system is verified on real-world operational datasets archived in the project repository:

- **3,980+ Clean Standardized Active Wash Telemetry Points:** Formatted in `data/pdm_ml_standardized_training_dataset.csv`.
- **5 Genuine Archived Wash Cycles:** Multi-minute and 100-minute cycles including `wash_cycle_2026-10-12_1019.csv` (1,216 records), `wash_cycle_2026-10-15_morning.csv` (1,216 records), and `wash_cycle_2026-10-16_0857.csv` (859 records with live Tapo power). Bench & idle traces are segregated in `data/testing_data/`.
- **Solving the Spin-Cycle Paradox:** By fusing Samsung SmartThings ground-truth operational states (`washerJobState`), Apex dynamically conditions anomaly thresholds, eliminating false alarms during normal 1,400 RPM spin extraction.

Category	Features	Physical Units	Diagnostic Significance
1. Kinematics	accel_x, accel_y, accel_z, vibe_mag	m/s^2	Chassis rocking, damper shock, and drum spider fatigue.
2. Acoustics	audio_rms, audio_rms_db, anomaly_pct	Canonical RMS, dB	Bearing micro-spall, surface friction, and cavitation.
3. Electrical Power	power_w, voltage_v, current_ma, energy_wh	W, V, mA, Wh	Motor torque, heating element health, and parasitic drag.
4. Ground Truth	phase, machine_state, target_rpm, temp	Categorical, RPM, °C	Contextual baseline conditioning to prevent false alarms.
5. Prognostics	parasitic_drag_w, machine_health_index	W, % (0%–100%)	Dynamic laundry mass, bearing health, and RUL forecast.

6. Macro Township Infrastructure Scaling: Commercial Assets

While proven on a front-load washing machine, the commercial value for Sime Darby Property lies in retrofitting commercial and communal rotating infrastructure:

Township Asset Location	Target Equipment	Critical Failure Mode	Apex Solution
Masjid Elmina & Suraus	4–8 Central AC Chillers (15–30 kW)	Compressor bearing fatigue & Friday prayer breakdown.	Pre-emptive bearing swap during off-peak weekdays.
Elmina Lakeside Mall	15–30 Commercial Rooftop Chillers	Parasitic motor drag; massive tenant electricity bills.	Continuous parasitic wattage tracking & alert.
Community Clubhouses	4–6 Pool Booster Pumps & HVAC	Impeller cavitation & dry-run damage.	Acoustic cavitation RMS alert & auto-cutoff.
Township Water Stations	2–4 High-Head Drainage Pumps	Pump seizure during heavy monsoon floods.	Periodic vibration health testing prior to rains.

Quantified Financial & Sustainability ROI (25 Commercial Assets)

- **Baseline Consumption:** 20 kW × 10 hrs/day × 365 days = 73,000 kWh/year per compressor.
- **Parasitic Friction Drag (15%):** 10,950 kWh/year wasted per degraded unit.
- **Financial Waste (TNB Tariff B @ RM 0.509/kWh): RM 5,573 wasted/year** per degraded unit. Across 10 degraded compressors, that is **> RM 55,700 in wasted electricity annually**.
- **Deployment Cost:** 25 pods × RM 170 = RM 4,250. **Payback period is under 1 month.**
- **Carbon Offset: 75 metric tons of CO&sub2; equivalent avoided annually** across 25 units.
- **Circular Economy: 25 to 40 metric tons** of compressor scrap metal and e-waste prevented.

7. Implementation Roadmap for Sime Darby Property

PHASE 1: MICRO TESTBED (COMPLETED & VERIFIED) • Edge rig operational on Samsung washer; Dual-Core FreeRTOS C++ firmware running. • 5,530+ verified operational points; Live Cloudflare dashboard at https://washer-pdm.pages.dev. ↓ PHASE 2: TOWNSHIP PILOT DEPLOYMENT (MONTHS 1–3) • Deploy 25 sensor pods across Masjid Elmina, Elmina Lakeside Mall, and Clubhouses. • Establish baseline vibration and power profiles under commercial loads. ↓ PHASE 3: PLATFORM & RESIDENT APP INTEGRATION (MONTHS 4–6) • Centralized Facility Management Command Dashboard for Sime Darby Property FM teams. • Integration with Sime Darby Township Resident App for home appliance health notifications. ↓ PHASE 4: TOWNSHIP-WIDE STANDARDIZATION (MONTHS 7–12) • Standard inclusion in future residential handovers across Bandar Bukit Raja and City of Elmina. • Automated circular parts replacement and ESG sustainability reporting.

Part 2: Official Kolaxus 6-Axis Evaluation Rubric Breakdown

Evaluation Axis	Weight	How Apex PdM Achieves Maximum Score	Submission Evidence
1. Impact & Sustainability Value	**25%**	Dual sustainability: eliminates 15%–35% parasitic electricity waste and extends asset lifespan by 5–10 years to prevent metric tons of e-waste.	TNB tariff equations; Peninsular Malaysia grid emission factor (0.758 kg CO&sub2;/kWh).
2. Innovation & Differentiation	**20%**	Novel multimodal fusion (50 Hz kinematics + 16 kHz DMA acoustics + 1 Hz power + SmartThings state). Phase-conditioned anomaly detection eliminates false alarms.	31-column ML contract in `pdm_feature_dictionary.json`; FreeRTOS dual-core code.
3. Feasibility & Deliverability	**20%**	Sub-RM 50 production BOM, non-intrusive clip-on magnetic/adhesive pods requiring zero rewiring, with automated Force-ON safety guards.	Turnkey BOM under RM 171 (< RM 50 at volume); physical prototype rig, IP65 enclosure photos, and engineering cutaways.
4. Strategic Fit & Theme Relevance	**15%**	Directly answers the "Township of Tomorrow" core brief (Value multiplying people, businesses, and planet). Anchored to City of Elmina assets.	Site-specific rollout for Masjid Elmina, Lakeside Mall, and retention ponds.
5. Commercial & Value Multiplication	**10%**	Scalable B2B SaaS and Facility Management model with an immediate payback period under 3 months for commercial buildings.	RM 55,700 annual electricity waste eliminated against RM 4,250 hardware cost.
6. Team, Evidence & Pitch Quality	**10%**	**Operating physical rig with working code**, 5,530+ verified real data points, and active live deployed dashboard accessible worldwide.	Live dashboard at `https://washer-pdm.pages.dev`; physical prototype photos; 6 real wash datasets.

Part 3: The 8-Slide Grand Finale Pitch Deck Blueprint

Slide 1: Cover & Vision — The Pulse of Sustainable Townships

SLIDE 1 (30s)

Visuals: High-tech split rendering: Physical IoT pod on left; township asset health dashboard on right. Live URL banner: `https://washer-pdm.pages.dev`.

> *"Judges, modern townships pride themselves on sustainability. Yet behind the scenes, hundreds of air-conditioning chillers, pumps, and household appliances run until they break down. We are Apex, and we are introducing multimodal predictive maintenance to eliminate friction, save electricity, and keep machines out of landfills."*

Slide 2: The Silent Crisis — Friction Leeching in Modern Townships

SLIDE 2 (40s)

Visuals: 3 waste callouts: Parasitic Drag (+15%–35% Watts), Catastrophic Breakdown (RM 25 fix vs RM 1,500+ motor burnout), and Landfill E-Waste.

> *"When a bearing loses lubrication, an inverter motor draws up to 35% excess electricity to overcome friction—undetected for months. When it finally locks up, a RM 25 component failure becomes a RM 1,500 motor replacement and 70 kg of landfill e-waste."*

Slide 3: Universal Electromechanical Physics & The P-F Interval

SLIDE 3 (40s)

Visuals: P-F Curve diagram showing Apex early detection at Point P vs traditional breakdown at Point F.

> *"Whether it is a 1,400 RPM washing machine or a 50 kW central chiller in Masjid Elmina, all rotating machinery fails identically: Micro-vibrations lead to acoustic friction, which causes parasitic electrical drag, and finally catastrophic breakdown. We catch it at Point P."*

Slide 4: The Apex Platform Architecture

SLIDE 4 (45s)

Visuals: Clean hardware diagram showing ESP32 Dual-Core separation, Tapo P110 power, and cloud pipeline.

> *"Our hardware costs under RM 50 in production. Using dual-core FreeRTOS processing, Core 1 handles 50 Hz vibration and 16 kHz audio without jitter, while Core 0 manages power monitoring and cloud sync. Our 31-feature ML contract conditions anomalies on machine state, eliminating false alarms."*

Slide 5: The Working Prototype & Live Proof (Micro-Testbed)

SLIDE 5 (45s)

Visuals: The "Trio of Empirical Proof": (1) Photo of Physical IP65 Prototype Rig, (2) 3D Cutaway Installation Diagram, and (3) Live 4-Card Dashboard HUD at `https://washer-pdm.pages.dev`.

> *"Judges, most hackathon submissions remain hypothetical slides. We physically built our prototype and stress-tested it on an active 1,400 RPM Samsung washer. Why? Because a washing machine drum with shifting wet laundry is one of the most violent, acoustically noisy electromechanical systems in existence. We proved that our sub-RM 50 edge sensor pod captures, packages, and streams vibration, acoustics, and power in real time with zero false alarms. Having proven this in the toughest testbed, mass-producing these pods for stationary AC compressors at Masjid Elmina or Elmina Lakeside Mall is mechanically simpler, low-risk, and immediately viable."*

Slide 6: Scaling to Macro Township Infrastructure

SLIDE 6 (40s)

Visuals: Map of City of Elmina showing 25 communal asset retrofits (Masjid Elmina, Lakeside Mall, Clubhouses).

> *"We are taking this from 1 washing machine to 25 commercial assets across City of Elmina. Facility managers get an automated health command center, while residents get transparent appliance health updates on the Sime Darby Resident App."*

Slide 7: Measurable Sustainability & Financial ROI

SLIDE 7 (30s)

Visuals: Metrics card: RM 55,700/year electricity saved, < 1 month payback, 75 tons CO&sub2; offset annually.

> *"Across 25 commercial chillers, eliminating 15% parasitic friction saves over 100,000 kWh and RM 55,000 annually. Against a hardware cost of RM 4,250, the capital payback is under one month, while cutting 75 tons of carbon emissions."*

Slide 8: Township Pilot Roadmap & Future Vision

SLIDE 8 (30s)

Visuals: 3-phase timeline: Pilot (Months 1-3), Integration (Months 4-6), Standardization (Months 7-12).

> *"Sustainable townships start with smart, friction-free engineering. With Apex, Sime Darby Property can lead Southeast Asia in circular, energy-efficient asset management. Thank you."*

Part 4: Rapid Online Portal Submission Q&A Pack

Q1: Problem Description (160 words)

In modern sustainable townships, electromechanical equipment—from residential washing machines to central air conditioning compressors in community mosques and retail centers—is managed reactively. When bearings, seals, or dampers degrade, they create mechanical friction. Because modern inverter motors automatically draw excess electrical current to maintain speed, degraded machinery leeches **15% to 35% excess electrical power** for months without tripping circuit breakers.

Eventually, unaddressed friction leads to total motor burnout or compressor seizure. A minor, unaddressed RM 25 component failure cascades into catastrophic hardware destruction (costing RM 800–RM 1,800+ for appliances and RM 15,000–RM 50,000+ for commercial chillers). This forces premature scrapping of 70 kg domestic machines and heavy commercial castings into landfills. In large developments, manual technician inspections miss early acoustic and kinematic anomalies, leaving township operators blind to mounting energy waste and sudden public service failures.

Q2: Innovation & Uniqueness (180 words)

Apex PdM is the first low-cost (< RM 50 production BOM) multimodal edge-AI platform that fuses four synchronized physical dimensions: **high-frequency 3-axis kinematics (±16g @ 50 Hz)**, **ultrasonic/acoustic sound pressure (16 kHz 24-bit I2S DMA)**, **active electrical power (W, V, mA @ 1 Hz)**, and **ground-truth operational phase**.

Existing commercial predictive maintenance systems cost thousands of ringgit per sensor, target only heavy industrial oil-and-gas plants, and produce frequent false alarms during normal high-speed operations. Apex solves this via **phase-conditioned anomaly modeling**: by conditioning vibration and sound against ground-truth operational states, our algorithm distinguishes between high-speed centrifugal dynamics and actual mechanical failure.

Furthermore, our FreeRTOS dual-core firmware architecture isolates 50 Hz sensor sampling from network streaming, ensuring zero dropped audio samples and millisecond-level responsiveness. Combined with our automated Force-ON safety guard, Apex delivers industrial-grade prognostics at a consumer-accessible price point.

Q3: Technical Implementation (185 words)

The physical prototype utilizes an ESP32 dual-core microcontroller running custom C++ FreeRTOS firmware: **Core 1** executes uninterrupted 50 Hz MPU-6050 I2C burst reading and 16 kHz INMP441 digital I2S DMA buffer capture, wrapping data into a standardized JSON telemetry envelope guarded by `portMUX_TYPE` spinlocks. **Core 0** executes dual-tier streaming: a 1 Hz hot path pushing to Google Firebase `/live.json` for live dashboard rendering, and a 5.0s batch archival path for ML model training, while concurrently polling Tapo P110 active power via encrypted local KLAP protocol. Ground-truth machine state is integrated via Samsung SmartThings REST API.

We overcame three critical technical hurdles: (1) Thread Starvation & Jitter solved by dual-core processor affinity and spinlock memory handoffs; (2) Spin Cycle False Alarms solved by a 31-feature ML prognostic contract conditioning vibration thresholds on ground-truth phase; (3) Smart Plug Safety solved by an automated background watchdog that detects accidental relay deactivation within 1,000ms and forces power back ON.

The system is validated by over **5,530 verified real operational data points** and an active live web dashboard deployed at `https://washer-pdm.pages.dev`.

Q4: Target Audience & Adoption (175 words)

Apex serves two primary user tiers across Sime Darby Property developments: (1) **Township Facility Managers (B2B)** managing commercial assets across community mosques, suraus, retail malls (e.g., Elmina Lakeside Mall), and clubhouse facilities. Adoption is driven by out-of-the-box retrofitting: non-intrusive clip-on pods attach to existing compressor housings in under 10 minutes without rewiring, feeding directly into a unified township facility dashboard. (2) **Township Residents (B2C)** in smart residential precincts, reached through native integration with the **Sime Darby Township Resident App** with intuitive Machine Health Index notifications.

To drive early adoption, Sime Darby Property can bundle Apex sensor pods as a standard smart sustainability amenity in new home handovers, while piloting commercial compressor monitoring across flagship communal buildings to demonstrate immediate utility bill reductions.

Q5: Feasibility & Execution Roadmap (180 words)

We have already completed Phase 1: a field-tested physical testbed on a Samsung front-load washer, dual-core FreeRTOS firmware, 5,530+ verified telemetry records, and a live cloud dashboard deployed on Cloudflare Pages.

Our future execution roadmap comprises three structured phases: (1) **Months 1–3 (Township Pilot):** Retrofit 25 communal rotating assets in **City of Elmina** (8 central AC chillers in Masjid Elmina, 11 rooftop compressors at Elmina Lakeside Mall, and 6 clubhouse filtration pumps). Validate energy waste reductions in live commercial operating conditions. (2) **Months 4–6 (Platform & App Integration):** Integrate machine health telemetry into the Sime Darby Resident Mobile App and deploy automated maintenance triage ticketing for certified local service providers. (3) **Months 7–12 (Township-Wide Scaling):** Standardize Apex pods as an optional sustainability add-on for new residential handovers across Bandar Bukit Raja and City of Elmina, establishing an automated circular economy parts-replacement network.